UNIVERSITY OF FREIBURG

Department of Computer Science

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Machine Learning Course
Summer Semester 2021
Retake Exam

Parts	Points
1) Regularization and Error Measures	/ 10
2) Decision Trees	/ 10
3) Neural Networks	/ 10
4) Ensembles (AdaBoost)	/ 10
5) Gradient-Boosted Decision Trees	/ 10
6) Hyperparameter Optimization	/ 10
Total	/ 60

This exam consists of 6 parts worth a total of **60 points** and you have **90 minutes** to answer the questions.

Please check the back of each page, the exam is printed on both sides.

This examination is closed book: **you are not allowed to use any notes**. Please put your name and matriculation number on every page.

Your name: _____ Table number: _____

Your matriculation number: _____

Your signature: _____

Good luck!

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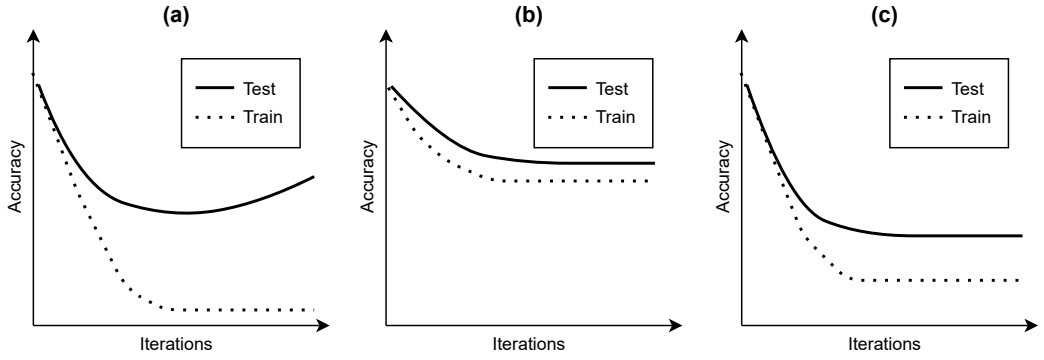
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Part 1 – Regularization and Error Measures (10P)

a. **(2P)** Fill in the blanks in the following sentence.

A Random Forest has a _____ bias and a _____ variance compared to a single unpruned (very deep) decision tree.

b. **(5P)** Consider training a neural network architecture on the same dataset with different DropOut drop rates (i.e. the probability of sampling a mask for each neuron) and Stochastic Gradient Descent, as shown in the a, b, c variants below.



Enter one of "a", "b", or "c" in the cells below.

- Analysis of the Bias-Variance Trade-off

Model Complexity Analysis	**"a", "b", or "c"?**
Low bias / Low variance	
High bias / Low variance	
Low bias / High variance	

- Measures for the variants that do not exhibit already a "Low bias / Low variance" complexity:

Measure	**"a", "b", or "c"?**
Increase the drop rate	
Decrease the drop rate	

c. **(3P)** A binary classifier $\hat{y}$ estimates the ground-truth variable y as shown in the table below. Compute the Precision, Recall and F1 measure of this classifier on the data given in the table.

y	$\hat{y}$
1	1
0	1
1	1
0	0

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Part 2 – Decision Trees (10P)

a. **(2P)** Given the following formalism for the prediction of a decision tree given a feature vector x, following the notation of the lecture:

$$f(x, \theta) = \sum_{j=1}^{J} \hat{y}^{R_j} \ \mathbb{1}(x \in R_j)$$

What do the terms R_j and $\hat{y}$ represent?

b. Consider training a decision tree in the dataset below where the target variable is denoted by y, and a single feature by x.

x_1	x_2	y
1	1	Circle
1	4	Cross
3	2	Circle
5	5	Square

I. What is the initial impurity of the data? **(1P)**

II. How many decision splits do we have based on the features? **(1P)**

III. Compute the decision stump on the given dataset that maximizes Information Gain. Show the resulting decision stump and the related calculation. ($\log_2(\frac{2}{3}) = -0.585, \log_2(\frac{1}{3}) = -1.585$) **(4P)**

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c. **(2P)** Suppose that you are given a simple regression dataset with one feature x and one target y. At the same time you are given a 2-dimensional visualization of the stepwise function created by a trained decision tree (on the same regression dataset) as an x vs. y plot.

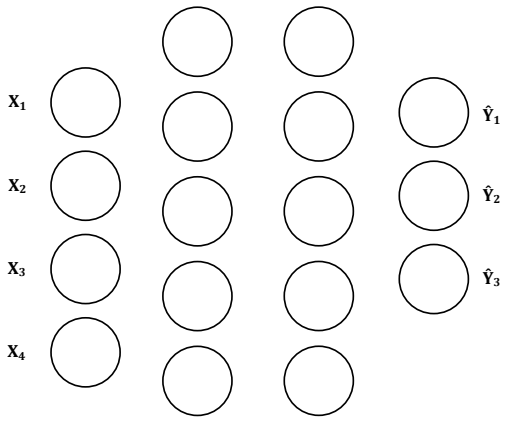
How can you spot the overfitting and underfitting of a decision tree, based on the visualization of the steps generated by the tree?

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Part 3 – Neural Networks (10P)

a. You are given the following feed-forward neural network structure:



I. **(1P)** How many weight layers does the network have?

II. **(1P)** How many weights does the given network have? (Bias should be included)

b. **(8P)** Consider a simplified neural network architecture, containing a single input feature x, a single estimated target $\hat{y}$, one hidden layer with ReLU activation ($f(x) = \max(0, x)$), weights w, and no bias terms.

$$\hat{y}(x; w) = w^{(2)} \text{ ReLU } \left(w^{(1)} x \right)$$

Your task is to optimize the neural architecture above for a ranking problem, where given a pair of data instances, then $r_{i,j} = 1$ indicates that $x^{(i)}$ has a higher rank than $x^{(j)}$, or vice versa. As a result, you will optimize the following logistic loss for every pair of instances $\left(x^{(i)}, x^{(j)}\right)$:

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$$\begin{aligned} \mathcal{L}_{i,j} &:= -r_{i,j} \ln\left(\sigma(\hat{r}_{i,j})\right) - (1 - r_{i,j}) \ln\left(1 - \sigma(\hat{r}_{i,j})\right) \\ &\text{s.t.} \quad \hat{r}_{i,j} := \hat{y}_i - \hat{y}_j := \hat{y}(x^{(i)}; w) - \hat{y}(x^{(j)}; w) \\ &\qquad \text{and } \sigma(x) = \frac{1}{1 + e^{-x}} \end{aligned}$$

We will use gradient descent to the update weights w for minimizing the loss $\mathcal{L}_{i,j}$. Show the back-propagation steps and derive the formula for the gradient $\frac{\partial \mathcal{L}_{i,j}}{\partial w^{(1)}}$.

Hint: $\frac{\partial \mathcal{L}_{i,j}}{\partial \sigma(\hat{r}_{i,j})} = -\frac{(r_{i,j} - \sigma(\hat{r}_{i,j}))}{\sigma(\hat{r}_{i,j})(1 - \sigma(\hat{r}_{i,j}))}$

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Part 4 – Ensembles (AdaBoost) (10P)

a. **(2P)** Show how the weights of ensemble models (α) are updated in AdaBoost, including how the error is computed. Denote terms and write down the formulae.

b. **(8P)** Instead of classification, you need to deploy the AdaBoost method to a regression problem with the least-squares error loss. What changes should you conduct to the components of AdaBoost to adapt it for regression? E.g.:

- How does the objective for training the regression trees change?
- What is the new rule for updating the data point weights?
- How would you compute the model coefficients?
- How would you compute the estimated target given a sequence of regression trees as ensemble components?

Write down very brief answers, e.g. of the form "[in comparison to the classification case] I will change X with Y in the formula of Z".

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(Write the solution of Q4 here if needed.)

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Part 5 – Gradient-Boosted Decision Trees (10P)

a. **(5P)** Consider optimizing the weights w of the current tree in a Gradient-Boosted Decision Tree ensemble (same as in the lecture). Let w_j denote the weight of the j-th leaf, G_n, H_n the gradient and hessian of the loss with respect to the estimated target variable of the current ensemble. Furthermore, the indices of data points belonging to the j-th leaf be defined as: $I_j := \{n \mid q(x_n) = j\}$. T denotes the number of leaves and λ, γ are regularization hyperparameters. We compute the optimal weights w^{opt}, that minimize the objective function $\mathcal{O}(w)$ by solving:

$$w^{\text{opt}} := \underset{w_1,\ldots,w_T}{\arg\min} \; \mathcal{O}(w) \; := \; \underset{w_1,\ldots,w_T}{\arg\min} \; \sum_{j=1}^{T} \left[\left(\sum_{n \in I_j} G_n \right) w_j + \frac{1}{2} \left(\lambda + \sum_{n \in I_j} H_n \right) w_j^2 \right] + \gamma T$$

Derive the formula of the minimum value of the objective for each leaf, i.e.: $\mathcal{O}\left(w_j^{\text{opt}}\right) =?$ as a function of G, H, λ, θ.

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b. **(5P)** The classification problem of the dataset in the table below contains 3 data points, where x is a single feature and y is a binary target. Build a single decision tree (an ensemble that contains a single tree) using the Gradient-Boosted Decision Tree technique with hyper-parameters $\gamma = \lambda = 0$. Assume the initial predictions before the first ensemble tree are $\hat{y} = 0$. Clearly and briefly show the intermediate calculations and the final decision tree.

No.	x	y
1	-1	1
2	0	0
3	1	1

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Part 6 – Hyperparameter Optimization (10P)

a. I. **(2P)** When an optimal hyperparameter configuration is needed to achieve peak performance, what stops a practitioner from using the response function, instead of using approximations to evaluate different hyperparameter configurations?

II. **(2P)** What is the benefit of using multi-fidelity hyperparameter optimization methods (HPO) compared to plain model-based/non-model-based methods?

b. **(3P)** You are given the following algorithm for sequential model based optimization (SMBO), where a stands for the acquisition function, f for the response function and Ψ for the surrogate function. Find the bugs in the algorithm and provide the correct implementation.

1. Initial observations $\mathcal{H} := (\lambda_1, f(\lambda_1, D)), ..., (\lambda_k, f(\lambda_k, D))$
2. **for** $t = 1$ to T **do**:
3. Fit Ψ to $\mathcal{H}$
4. $\lambda_{new} = \arg\max_{\lambda \in \Lambda} a(\lambda, f(\lambda))$
5. Evaluate $f(\lambda_{new}, D)$
6. $\mathcal{H} = \{(\lambda_{new}, f(\lambda_{new}, D))\}$
7. Return $\lambda^* = \arg\min_{\lambda \in \mathcal{H}} \Psi(\lambda, D)$

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c. **(3P)** You are given a choice between a non-model based HPO method (e.g. random search with no model or parameters) and a model based optimization HPO method (e.g. SMBO with Gaussian Processes). Explain at least one scenario where you would choose one over the other (i.e. at least two scenarios in total) and the reason behind your choice.